PRITHVI KRISHNASWAMY

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Engineering Skills: Altium Designer, LTSpice, Python, Cadence Virtuoso/Spectre, MATLAB, Simulink, PLECS, Ansys HFSS

Education Credentials

Bachelor of Science: *Electrical Engineering* | **University of Washington (UW)**, Seattle, WA | Graduation: 06/2025
Minor: *Applied Mathematics* | **Coursework**: *Circuit Theory, Energy Systems, Applied Electromagnetics* | **GPA:** *3.57*
Master of Science: *Electrical Engineering* | **University of Washington (UW)**, Seattle, WA | Graduation: 06/2027
Coursework: *Power Electronics, Control Systems, Analog Integrated Circuits, Antenna Design* | **GPA:** *3.82*

Professional Experience

Avionics Hardware Design Engineering Intern (Incoming) **| Stoke Space** | Kent, WA | Sep 2026 – Dec 2026
- Design & Testing **Servo Controllers** For Nova: Stoke Space's fully reusable launch vehicle!

EV Electrical Design Engineering Intern **| Rivian/Volkswagen Technologies** | Irvine, CA | Jun 2026 – Sep 2026
- Tuned reliability of Pierce Oscillator Circuit across 10 Zonal Controllers, 4 vehicle families & up to ~100,000 vehicles.
- Characterized 12V/48V HSDs & validated **RCP** Emulation for Zonal-Controlled edge nodes, advancing a SW-less architecture eliminating local MCUs, device-resident application software/OTA burden & reduced wiring complexity
- Normal-Load Efficiency/Thermal Performance characterization of a **2-Phase 48V-to-15V GaN DC-DC** Converter across 1-34 A loads, 36V/48V/54V inputs, and -40 C to 85 C test corners. Demonstrated 98.03% efficiency
- Tested & Updated a **48V-to-12V short-circuit protection** test board for 12V HSDs against 440 A transients under a 48 V Bus Fault. Compared component-level surge-energy, clamping voltage and recovery performance

Aircraft Electrical Design Engineering Intern **| Shield AI** | Dallas, TX | Jan 2026 – Jun 2026
- Hardware Bring-up, DVT & Qualification of V-BAT **ViDAR** **eXtensible Processing Module (XPM).** Validated Electrical**,** Thermal, OS & Mission Software Parameters for 1st stage Experimental Flight Test (XFT)
- Prototyped DAQ Interposer PCB to validate connector stability. Scripted **Python** vibration tests & logged data using **LabVIEW**
- Owned EE responsibilities for V-BAT **Battery Management System (BMS)** enabling cell-level Temperature, Voltage, State of Health (SOH), and State of Charge (SOC) monitoring. Added enhanced Protection and Sequence Mechanisms. Qualified Thermal, Vibration & Quiescent Current Discharge Performance to match design/operating parameters

Datacenter Hardware Design Engineering Intern **| Nvidia |** Durham, NC | Jun 2024 – Sep 2024
- Designed a simplified GPU-emulator PCB in **Cadence Allegro** that emulated key power, signal, and I²C interfaces of the **GB200** server switch tray, cutting test-fixture expenses by ≈ 85% and accelerating bring-up cycles.
- Qualified **Single/Multi Phase Buck & LDO rails** on **50xx GPU** SLT boards. Characterized FRA Bode Plot, Transient, Efficiency, Input Droop, Output Ripple, and AMR measurements to achieve ≥55° phase margin and <30 mVpp at rated load.
- Re-architected a Power Adapter PCB for 100+ engineers— transitioned to CEM5 high-current connectors, added OVP/UVP, inrush control - to improve safe & flexible AC/DC and DC/DC power-conversion for DGX/MGX server platforms.

Plasma Etch Silicon Engineering Intern **| Intel** | Hillsboro, OR | Jan 2023 – Dec 2023
- Innovated a custom **Deep Reactive Ion Etching (DRIE)** **Bosch Process**, achieving 50:1 to 100:1 aspect ratio trenches and holes for MEMs Through-Silicon Vias (TSVs). Enhanced etching precision by 3x baseline on 20 validated silicon wafers.

Automotive Technician & App Developer **| Singapore Armed Forces |** Singapore | Oct 2019 – July 2021
- Mechanical & Electrical maintenance on 4 different fleets of 20 vehicles. Conducted Failure Analysis, debugging & validation of Batteries, Filters & Drivetrains. Catalogued defects & ordered necessary replacements.
- Constructed a user-friendly front-end interface for a multiplatform application aimed at increasing workshop safety awareness; designed back-end connections that ensured secure handling of user input data for over 100 users
- Beta-tested the application with documentations and **training videos** that increased user adoption by a factor of 4

Projects

Electronics Engineer **| UW Formula Motorsports (FSAE) |** Seattle, WA | Jul 2025 – Jun 2026
- Architecting an **Integrated Powertrain Module** to reduce HV harness length, simplify packaging, and improve EMI behavior
- Spearheading architecture & design of a custom 900W, isolated **Resonant LLC DC – DC Converter** reducing battery costs by 90%, while achieving dual compatibility with 400V & 600V HV pack architectures
- Renovating Schematic & Layout, in **Altium**, of a custom 26 kVA, 540V DC high-performance **Quad – Motor Inverter** to decrease car weight by 11kg and increase efficiency up to ~98 %

Extracurriculars

Teaching Assistant **| UW Department of Electrical & Computer Engineering** | Seattle, WA | Jun 2023 – Jun 2025
- Built a **line-following robot platform** used by 100+ students, to teach full-stack prototyping (PCB → firmware → validation).
- Coordinated 100+ office-hour sessions (≈30 students/week) and streamlined grading for labs & HW

Lead Research Assistant **| Washington Nanofabrication Facility** | Seattle, WA | May 2022 - Current
- Validated Vision **Reactive Ion Etcher (RIE)** quality & reliability. Subjected ~100 silicon wafers across various processes, verifying trench depth & uniformity using a **Scanning Electron Microscope (SEM)**. Published process notes for lab users.
- Doubled etching rate of Oxford **Inductively Coupled Plasma (ICP) Etcher** for PECVD & Thermal Oxide deposited Silicon Dioxide wafers. Documented parameters for reproducibility.